USART

Universal Synchronous Asynchronous Receiver Transmitter

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OVERVIEW

Overview & Key Features

USART is a highly flexible serial communication peripheral capable of operating in multiple modes, making it a cornerstone of embedded system communication design.



Full-Duplex NRZ

Simultaneous transmit and receive using standard Non-Return-to-Zero asynchronous serial format.



Multi-Mode Support

Operates in synchronous, half-duplex, and multiprocessor networking configurations.



Protocol Compatibility

Compatible with LIN, Smartcard (ISO 7816), IrDA SIR ENDEC, and DMA-based data transfer.

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HARDWARE

Essential Pins

USART relies on a defined set of signal pins to manage data flow, clocking, and hardware handshaking. Each pin serves a distinct role in maintaining reliable communication.

SCLK — Clock Output

Provides a clock signal to the receiver in synchronous communication mode only.

nCTS — Clear to Send

Hardware flow control input. Pauses transmission when asserted by the receiving device.

nRTS — Request to Send

Hardware flow control output. Signals readiness to receive data from the remote device.

TX — Transmit Data Output

Transmits serial data outward. Held at logic high (idle state) when no transmission is active.

RX — Receive Data Input

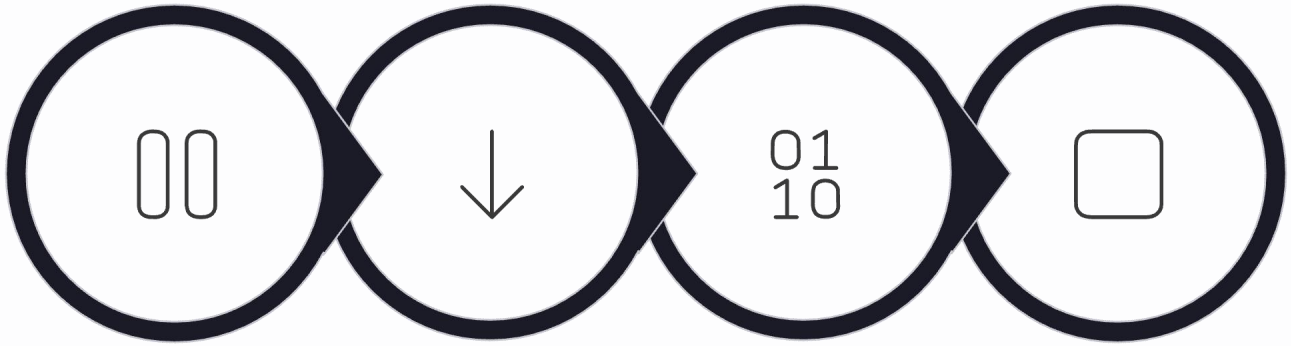
Accepts incoming serial data. Employs oversampling noise filtering to ensure signal integrity.

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FRAME STRUCTURE

Character Frame Structure

Every USART transmission follows a defined frame structure. Understanding each field is essential for correct configuration and reliable data exchange.



Idle Line | Start Bit | Data Word | Stop Bits

Idle Line

The line is held at logic "1" before any transmission begins, signaling an inactive but ready state.

Start Bit

A single logic "0" marks the beginning of each character frame, synchronizing the receiver.

Data Word

8 or 9 data bits transmitted LSB (Least Significant Bit) first, configurable via the M bit in USART_CR1.

Stop Bits

Configurable stop duration: 0.5, 1, 1.5, or 2 bits, providing inter-frame separation and synchronization.

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TRANSMITTER

Transmitter Operation

The USART transmitter follows a precise register-driven sequence. Each step must be completed in order to ensure correct and complete frame transmission.



Enable TE Bit

Set the **TE** bit in `USART_CR1` to activate the transmitter and prepare the idle line.



Write to DR

Write data to `USART_DR`. It is automatically moved to the Transmit Shift Register (TSR).



Monitor TXE

The **TXE** flag goes high when the TDR is empty and ready to accept the next data byte.



Confirm TC

The **TC** flag is set once the full frame including stop bits has been completely shifted out.

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RECEIVER & BAUD RATE

Receiver Operation & Baud Rate

Receiver Operation

The USART receiver continuously monitors the RX line for a valid start bit. Once detected, it samples incoming bits using either **8× or 16× oversampling** to filter noise and reconstruct the data word accurately. Upon successful reception of a complete character frame, the **RXNE** (Read Data Register Not Empty) flag is set, signaling the application to read the received byte from `USART_DR`.

Hardware Flow Control

The **RTS/CTS** mechanism prevents receiver overrun and buffer overflow. RTS signals readiness to receive; CTS pauses the transmitter when the peer is not ready — critical in high-throughput applications.

Baud Rate Configuration

The baud rate is derived from the peripheral clock using fractional division via the `USARTDIV` register.

Fractional baud rate generation minimizes clock error across standard baud rates such as 9600, 115200, and beyond.

12-bit Mantissa

Integer portion of the clock divisor value.

3/4-bit Fraction

Fractional portion enabling precise baud rate tuning.

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Thank You

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